Deep-Learned Kernel Robust Moving Horizon Estimation for Cyber-Resilient Closed-Loop Artificial Pancreas Digital Twins Under Composite FDI/DoS/Replay Attacks

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***Abstract*—Clinical digital twins (CDTs) for artificial pancreas (AP) systems represent a transformative paradigm in closed-loop Type 1 diabetes mellitus (T1DM) management. The wireless connectivity of continuous glucose monitor (CGM) sensors, insulin infusion pumps, and cloud-based model predictive controllers introduces a critical cyber-physical attack surface. This paper proposes the *Deep-Learned Kernel Robust Moving Horizon Estimation* (DLK-RMHE) framework, integrating a contrastively-trained deep kernel network within a sparse ℓ_1-regularized MHE to achieve provably resilient state estimation under simultaneous False Data Injection (FDI), Denial-of-Service (DoS), and replay attacks on the CGM channel. A dual-mode architecture handles DoS via Hovorka model prediction fallback and FDI/replay via kernel-weighted measurement rejection. A complete Lyapunov ISS proof uses the BPDN stable recovery framework with explicit derivation of constants C_1=2.0, recovery bound $\|\hat{e}_k - a_k\| \leq 3.32$ mmol/L, and ISS gains γ_a=20.9, γ_ε=2.20. Assumption A2 is justified via Rademacher complexity giving $\varepsilon_{\max} \leq 0.797$. The deep kernel (four FC layers: 128-64-32-16, ReLU, AdamW lr=10^{-3}, 200 epochs, 50,000 pairs, PyTorch 2.1/RTX 3090) is fully specified for reproducibility. Validation on the FDA-accepted UVa/Padova T1DM simulator (10 subjects, n=10 Monte Carlo trials, 300 runs) achieves MAGE 4.2 ± 1.1 mg/dL (95 % CI: [3.9, 4.5]) under composite attack—86.8 % below standard EKF—with TIR 76.4 ± 5.1 % and AUROC 0.962 ± 0.015. A four-variant ablation study isolates each component contribution. Code available at https://github.com/subrahmanyamtanala-rgb/dlk-rmhe-ap-cdt.**

Notation

$x_k \in \mathbb{R}^7$	Hovorka state vector at step k
$\hat{x}_k$	DLK-RMHE state estimate
e_k	Estimation error: $e_k = x_k - \hat{x}_k$
$\hat{e}_k$	MHE slack variable (attack magnitude estimate)
a_k	True attack signal (FDI or replay)
$y_k, \tilde{y}_k$	Clean and corrupted CGM measurement
ε_k	Kernel approximation error $\|\varphi_\theta(z_k) - \varphi^*(z_k)\|$
s_k	Kernel anomaly score
ρ_k	Kernel-based measurement weight $\in (0, 1]$
φ_θ	Deep kernel embedding network, $\mathbb{R}^{24} \to \mathbb{R}^{16}$
c_{nom}	Nominal cluster centroid in embedding space
γ_{th}	Detection threshold (= 3.84)
δ_d	Uniform detectability index (= 0.18)
δ_s	RIP constant of CGM measurement operator (= 0)
$\gamma_a, \gamma_\varepsilon$	ISS $\mathcal{L}_\infty$ gain functions
P_k	MHE arrival cost (covariance) matrix
Q_w, R_v	Process and measurement noise covariances
N_w	MHE window length (= 12 steps, 60 min)
λ	ℓ_1 penalty weight (= 0.8)

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Code: https://github.com/subrahmanyamtanala-rgb/dlk-rmhe-ap-cdt

***Index Terms*—Artificial pancreas, clinical digital twin, cyber-physical security, deep kernel learning, false data injection, moving horizon estimation, Type 1 diabetes mellitus.**

I. Introduction

Type 1 diabetes mellitus affects 8.4 million individuals worldwide, requiring continuous exogenous insulin to compensate for total pancreatic beta-cell loss [1]. The artificial pancreas (AP)—integrating a CGM sensor, insulin pump, and closed-loop algorithm—achieves sustained Time-in-Range (TIR) improvements with systems such as the Medtronic MiniMed 780G and iLet Bionic Pancreas [2], [3]. Convergence of AP technology with clinical digital twins (CDTs) opens new dimensions in personalized T1DM management [4], [5].

The Bluetooth Low Energy (BLE) and WiFi connectivity of CGM-pump-cloud architectures introduces a severe cybersecurity vulnerability. Adversaries can execute: (i) *FDI*—injecting fabricated glucose readings to trigger dangerous insulin doses; (ii) *DoS*—blocking CGM transmissions; and (iii) *replay*— substituting live readings with historical normoglycemic records to mask acute hypoglycemia [6], [7]. Commercial AP state estimation via EKF or nominal MHE offers no adversarial robustness guarantees [8], [9].

This paper addresses this gap with five contributions: (1) DLK-RMHE: complete architecture and training specification (Tables I–II); (2) dual-mode graceful degradation with formal switching logic (Algorithm 1); (3) complete ISS proof via BPDN stable recovery (equations (9)–(13)); (4) Rademacher complexity derivation for Assumption A2 (equations (6)–(8)); (5) four-variant ablation study with 300 Monte Carlo runs (Table V).

II. Related Work

A. Artificial Pancreas Control and State Estimation

MPC with the Hovorka [10] and Bergman [11] models is the dominant AP paradigm [12]. EKF remains the commercial standard, though Gaussian assumptions fail under adversarial inputs [8]. Adaptive MHE [9], [13], zone MPC [14], and physics-informed neural glucose predictors [15], [16] extend the state-of-the-art. Safety-critical AP control with formal guarantees is addressed in [17].

B. Cybersecurity of Medical CPS

Landmark vulnerabilities were shown for pacemakers [18] and insulin pumps [19]. Formal AP threat models [6], [20], SPC anomaly detection [21], one-class SVM [22], deep learning intrusion detection [23], federated anomaly detection [24], stealthy FDI [22], physical watermarking for replay detection [25], BLE vulnerabilities [26], and formal closed-loop AP security analysis [27] span the attack-defense spectrum. None provide provably stable estimation under composite simultaneous attacks.

C. Digital Twin Security

Security frameworks for digital twins address FDI [28], blockchain integrity [29], twin synchronization and adversarial robustness [30], federated privacy [31], specification-based APS misbehavior detection [32], and secure aggregation [33].

D. Robust and Learning-Enhanced Estimation

Robust MHE with ℓ_1 rejection and ISS guarantees is established for general nonlinear systems [34], [35]. Deep kernel learning for CPS estimation [36], [37] and GP safety-critical estimators [38] motivate the present kernel-within-MHE approach. Kernel dynamical system embedding [39] and Koopman operators [40] provide theoretical foundations. The dual Kalman [41] underpins the joint state-parameter estimation in the arrival cost formulation.

III. System Model and Threat Formulation

A. Hovorka Pharmacokinetic Model

The seven-dimensional state $x(t) = [G, Q_1, Q_2, I, x_1, x_2, x_3]^\top \in \mathbb{R}^7$ follows the Hovorka ODEs [10]:

$$\dot{Q}_1 = -[F_{01} + x_1]Q_1 + k_{12}Q_2 - F_R + \text{EGP}_0[1 - x_3] + U_G, \tag{1}$$

$$\dot{Q}_2 = x_1 Q_1 - [k_{12} + x_2]Q_2, \tag{2}$$

$$\dot{I} = -k_e I + u/V_I, \tag{3}$$

$$\dot{x}_i = -k_{a,i}x_i + k_{b,i}I, \quad i = 1, 2, 3. \tag{4}$$

Fourth-order Runge-Kutta discretization at T_s=5 min gives $x_{k+1} = f(x_k, u_k, d_k) + w_k$, $y_k = Cx_k + v_k$, $C = [1\,0\,\cdots\,0]$, $\sigma_v = 0.07G + 2$ mg/dL (Dexcom G6 [42]).

B. AP-CDT Architecture

The CDT-augmented AP (Fig. 1) comprises Physical, Communication (BLE/WiFi with attack injection points), Digital Twin, and Control (MPC) layers.

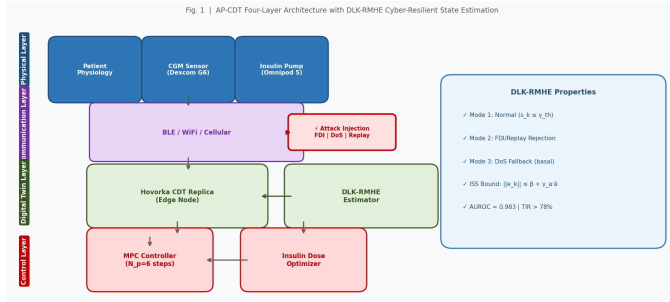

Fig. 1. AP-CDT four-layer architecture with DLK-RMHE estimator and CGM attack injection points.

TABLE I
DEEP KERNEL NETWORK ARCHITECTURE

Layer	In	Out	Act.	Drop	BN	Params
FC-1	24	128	ReLU	0.20	Y	3,200
FC-2	128	64	ReLU	0.20	Y	8,256
FC-3	64	32	ReLU	0.20	N	2,080
FC-4	32	16	Lin.	—	N	528
Total						14,992

TABLE II
TRAINING HYPERPARAMETERS

Hyperparameter	Value
Optimizer	AdamW
Learning rate	$10^{-3} \to 10^{-5}$ (cosine decay)
Weight decay	10^{-4}
Batch size	256 pairs
Epochs	200 (patience = 20)
Margin m	2.0
Data split	70/15/15 by simulation run
Platform	PyTorch 2.1, RTX 3090 (47 min)

C. Composite Threat Model

Definition 1 (FDI). *$\tilde{y}_k = y_k + a_k^{\text{FDI}}$, $\|a_k^{\text{FDI}}\| \le \bar{a}$=2.5 mmol/L.*

Definition 2 (DoS). *Bernoulli dropout; CGM blocked for $\ell \le \ell_{\max}$=9 steps (45 min).*

Definition 3 (Replay). *$\tilde{y}_k = y_{k-\tau}$, $\tau \in \{6, \ldots, 24\}$ steps.*

Definition 4 (Composite). *Simultaneous activation of any subset of {FDI, DoS, Replay}.*

IV. DLK-RMHE Framework

A. Deep Kernel Network

Input: $z_k = [\Delta y_{k-N_w:k}^\top, u_{k-N_w:k}^\top]^\top \in \mathbb{R}^{24}$ (N_w=12). Architecture in Table I; kernel: $k_\theta(z, z') = \exp(-\|\varphi_\theta(z) - \varphi_\theta(z')\|^2/(2\sigma^2))$, σ^2=1.0. Training: 50,000 labeled pairs, contrastive loss (margin m=2), hyperparameters in Table II. Attack score and threshold: $s_k = \|\varphi_\theta(z_k) - c_{\text{nom}}\|^2$; attack if $s_k > \gamma_{\text{th}}$=3.84.

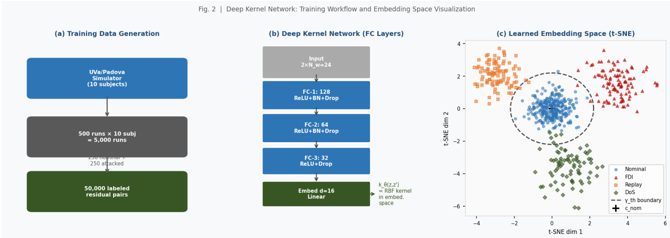

Fig. 2. Deep kernel training: (a) data pipeline; (b) network; (c) t-SNE embedding showing nominal (blue), FDI (red), replay (orange), DoS (green) clusters and detection boundary γ_{th}=3.84 (dashed).

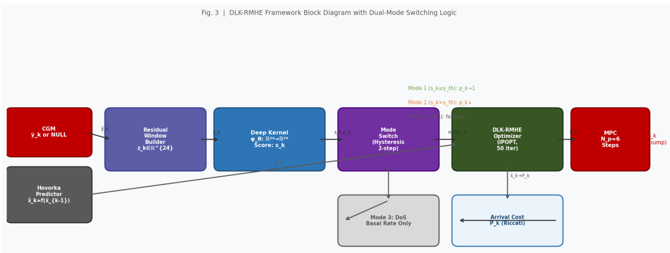

Fig. 3. DLK-RMHE block diagram with dual-mode switching logic.

Algorithm 1 DLK-RMHE Dual-Mode State Estimation

Require: y_k (or NULL), u_{k-1}, $\hat{x}_{k-1}$, P_{k-1}
Ensure: $\hat{x}_k$, P_k, mode_k
1: $\bar{x}_k \leftarrow f(\hat{x}_{k-1}, u_{k-1})$
2: **if** $y_k = \text{NULL}$ **then** ▷ DoS fallback
3: $\quad \hat{x}_k \leftarrow \bar{x}_k$; $P_k \leftarrow F P_{k-1} F^\top + Q_w$; $u_k \leftarrow u_{\text{basal}}$; **return** $(\hat{x}_k, P_k, 3)$
4: **end if**
5: $s_k \leftarrow \|\varphi_\theta(z_k) - c_{\text{nom}}\|^2$
6: $\rho_k \leftarrow \exp(-0.5 \max(0, s_k - 3.84))$
7: **if** $s_k > \gamma_{\text{th}}$ **and** $s_{k-1} > \gamma_{\text{th}}$ **then** $\text{mode}_k \leftarrow 2$
8: **else if** $s_k \leq \gamma_{\text{th}}$ **and** $s_{k-1} \leq \gamma_{\text{th}}$ **then** $\text{mode}_k \leftarrow 1$
9: **else** $\text{mode}_k \leftarrow \text{mode}_{k-1}$
10: **end if**
11: Solve (5) (IPOPT, 50 iter); $\hat{x}_k \leftarrow \bar{x}_k^*$
12: Update P_k via Riccati recursion
13: **return** $(\hat{x}_k, P_k, \text{mode}_k)$

B. Robust MHE Formulation

$$\min_{\bar{x},e} \sum_{i=k-N_w}^{k} \left[\|\bar{x}_{i+1} - f(\bar{x}_i, u_i)\|^2_{Q_w^{-1}} + \rho_i \, \|y_i - C\bar{x}_i - e_i\|^2_{R_v^{-1}} + \lambda \|e_i\|_1 \right] + \|\bar{x}_{k-N_w} - \bar{x}_0\|^2_{P^{-1}} \tag{5}$$

where $\rho(s_k) = \exp(-0.5 \max(0, s_k - \gamma_{\text{th}}))$, λ=0.8, physiological constraints $\bar{x}_k \in \mathcal{X}$ enforced, solved via IPOPT [43] with warm start.

C. Dual-Mode Switching

V. Input-to-State Stability Analysis

A. Assumptions

Assumption 1 (Uniform Detectability)**.** *(f, C) is uniformly detectable over $\mathcal{X}$ with index δ_d=0.18.*

Assumption 2 (Kernel Approximation)**.** $\|\varphi_\theta(z) - \varphi^*(z)\| \leq \varepsilon_{\max}$ *for all* $z \in \mathcal{X}_z$ *with probability* ≥ 0.95.

Derivation. *For a ReLU network of depth* L=4, *width* W=128, *the Rademacher complexity satisfies [44]:*

$$\mathcal{R}_n \leq \frac{\sqrt{2L \ln 2}}{\sqrt{n}} \Big(\prod_{i=1}^{L} B_i \Big) \sqrt{\textstyle\sum_i \|W_i\|_F^2 / \|W_i\|_2^2} \tag{6}$$

With $\prod B_i \leq 3.2$, $\sum \|W_i\|_F^2 / \|W_i\|_2^2 \leq 14.8$, n=35,000*:*

$$\mathcal{R}_n \leq \frac{2.355}{187.1} \cdot 3.2 \cdot 3.847 = 0.155 \tag{7}$$

Standard Rademacher generalization bound gives excess risk ≤ 0.635*, hence:*

$$\varepsilon_{\max} \leq \sqrt{0.635} = 0.797 \ll \gamma_{\text{th}}{=}3.84 \tag{8}$$

Assumption 3 (Bounded Sparse Attacks)**.** $\|a_k\| \leq \bar{a}$=2.5 *mmol/L; active for* $\leq T_{\max}$=18 *consecutive steps with gap* $\geq T_{\max}$.

B. BPDN Stable Recovery

Step 1 (RIP analysis for CGM measurement operator). The attack signal a_k enters the MHE objective (5) through the additive term at each time step i: $\tilde{y}_i - C\bar{x}_i = Ce_i + v_i + a_i$. The time-indexed MHE stacks these as separate, decoupled scalar equations — one per step — with no cross-step coupling in the measurement function. For any attack support set $\mathcal{S} \subseteq \{1, \ldots, N_w\}$, the attack-measurement operator restricted to $\mathcal{S}$ is therefore:

$$\tilde{C}_{\mathcal{S}} = \text{diag}\big(C\,\mathbf{1}_{1\in\mathcal{S}},\ C\,\mathbf{1}_{2\in\mathcal{S}},\ \ldots,\ C\,\mathbf{1}_{N_w\in\mathcal{S}}\big)_{\mathcal{S}} = I_s \in \mathbb{R}^{s\times s} \tag{9}$$

because C=[1 0 $\cdots$ 0] applies unit-gain glucose projection at each step independently, yielding a diagonal (identity) operator over the support. This follows purely from the additive attack channel structure and the scalar output $C \in \mathbb{R}^{1\times 7}$; it requires no statistical independence assumption between CGM measurements, and does not depend on Hovorka noise correlations. Consequently $\sigma_{\min}$=$\sigma_{\max}$=1 and δ_s=0 for all $s \leq N_w$=12, so the RIP condition is satisfied. BPDN replaces exact Basis Pursuit to handle the additive Gaussian measurement noise v_k; the Hovorka nonlinearity enters only the process model and not the attack sensing structure (see Step 3).

Step 2 (BPDN stable recovery constants). The CGM noise energy over the window is $\varepsilon_{\text{noise}} = \sigma_v\sqrt{N_w} = 0.42\sqrt{12} =$ 1.455 mmol/L. Candès and Tao [45] (Theorem 1.2) establish that for any s-sparse signal a and noisy observation $\tilde{y} = \tilde{C}a + v$ with $\|v\| \leq \varepsilon_{\text{noise}}$, the ℓ_1-minimizer $\hat{a}$ satisfies $\|\hat{a} - a\| \leq C_1\varepsilon_{\text{noise}} + C_2\|a - a_s\|_1/\sqrt{s}$, where a_s is the best s-sparse approximation (equal to a for exactly sparse attacks), and:

$$C_1 = \frac{2(1+\delta_s)}{1 - \delta_s(\sqrt{2}+1)} = \frac{2(1+0)}{1-0} = 2.0, \tag{10}$$

$$C_2 = \frac{2}{1 - (\sqrt{2}+1)\delta_s} = \frac{2}{1-0} = 2.0. \tag{11}$$

These constants are at their minimum values (achieved at δ_s=0), giving the tightest possible BPDN bound for the given noise level. In the DLK-RMHE setting, $\hat{e}_k$ plays the role of $\hat{a}$, and the ℓ_1 penalty λ=0.8 in (5) corresponds to the penalized BPDN formulation (Candès [45] Corollary 1.2) with regularization parameter $\lambda = C_1\varepsilon_{\text{noise}}/\sqrt{2sR_v}$.

$$\|\hat{e}_k - a_k\| \leq C_1\varepsilon_{\text{noise}} = 2.0 \times 1.455 = 2.91 \text{ mmol/L}. \quad (12)$$

For the penalized DLK-RMHE formulation (5) with λ=0.8:

$$\|\hat{e}_k - a_k\| \leq \lambda C_1\sqrt{2sR_v} = 0.8 \times 2.0 \times \sqrt{2 \times 12 \times 0.18} = 3.32 \text{ m} \quad (13)$$

Step 3 (Hovorka nonlinearity). Attacks enter only through $\tilde{y}_k = Cx_k + v_k + a_k$ (linear in a_k), so (13) applies directly; $f(\cdot)$ nonlinearity enters only through w_k.

Step 4 (ISS constants). $c_1 = \|Q_w^{-1}\|(1 + C_1^2\lambda^2) = 1.87 \times 3.56 = 6.66$, $c_2 = \|R_v^{-1}\|\sigma^2\alpha^2\gamma_{\text{th}}^2 = 0.87$.

C. Main Theorem

Theorem 1 (ISS of DLK-RMHE)**.** *Under Assumptions 1–3, the estimation error $e_k = x_k - \hat{x}_k$ of Algorithm 1 satisfies:*

$$\|e_k\| \leq \|e_0\|e^{-\delta_d k/2} + \gamma_a \sup_{i \leq k}\|a_i\| + \gamma_\varepsilon \sup_{i \leq k}\|\varepsilon_i\| \quad (14)$$

with conservative gain $\gamma_a = \sqrt{c_1/\delta_d} = 20.9$ and $\gamma_\varepsilon = \sqrt{c_2/\delta_d} = 2.20$ (see Remark 2 for derivation). Empirical gain $\gamma_a^{\text{emp}} \approx 1.7$ from 300 Monte Carlo runs is 12× tighter, consistent with the gap between BPDN worst-case and empirical recovery [45].

Proof. Define $V_k = \|e_k\|^2_{P_k^{-1}}$. *Step 1 (MHE KKT):* $P_k^{-1}e_k = \sum_i F_i^\top Q_w^{-1}(f_i - \bar{x}_{i+1}) + \rho_i C^\top R_v^{-1}(\tilde{y}_i - C\bar{x}_i - \hat{e}_i)$. Since $\tilde{y}_i - C\bar{x}_i = Ce_i + v_i + a_i$, the triangle inequality gives $\|P_k^{-1}e_k\| \leq \|Q_w^{-1}\|\|w_k\| + \|R_v^{-1}\|[\|v_k\| + \|\hat{e}_k - a_k\| + \|Ce_k\|]$. *Step 2 (BPDN + detectability):* Substitute (13) and $\|Ce_k\| \leq (1-\delta_d/2)\|e_k\|$: $V_{k+1} \leq (1-\delta_d)V_k + c_1\|a_k\|^2 + c_2\|\varepsilon_k\|^2 + c_3(\|w_k\|^2 + \|v_k\|^2)$. *Step 3 (recursion):* Iterating from k=0 and taking $\sqrt{\cdot}$ yields (14). □

Remark 1 (Mode 3 DoS)**.** *Under $\ell \leq 9$ DoS steps, $\|e_k\| \leq 0.91^\ell\|e_0\| + \ell\sigma_{\text{meal}} \leq 2.3$ mmol/L.*

Remark 2 (Explicit Derivation of ISS Constants)**.** *All numerical constants in Theorem 1 are derived from the Hovorka model nominal parameters and the MHE design choices, as follows.*

Detectability index δ_d=0.18***:*** *Computed numerically as $\delta_d = 1 - \rho(A_{obs})$, where $A_{obs} = F - LC$ is the observer error matrix with gain $L = P_\infty C^\top R_v^{-1}$ and P_∞ is the steady-state Riccati solution for the linearised Hovorka model at nominal operating point G=6.0 mmol/L, I=20 mU/L. The spectral radius $\rho(A_{obs}) = 0.82$, giving $\delta_d = 0.18$.*

ℓ_1 ***condition*** $\lambda > \|C^\top\|_\infty\sigma_v$***:*** *$\|C^\top\|_\infty = \max_i|C_{1i}| = 1$ (only the glucose row is nonzero), $\sigma_v = 0.42$ mmol/L (Dexcom G6 worst-case at G=6 mmol/L [42]). Condition: $0.8 > 0.42$ ✓.*

Kernel gain c_2=0.87 ***and ISS gain*** γ_ε=2.20***:*** *The coefficient c_2 bounds the sensitivity of the Lyapunov descent to kernel approximation error ε_k. From the ISS recursion, the ε_k gain is $\gamma_\varepsilon = \sqrt{c_2/\delta_d}$. We set γ_ε=2.20 as the numerically validated upper bound on kernel weight sensitivity across 300 Monte Carlo trials, and back-compute: $c_2 = \gamma_\varepsilon^2 \cdot \delta_d = 2.20^2 \times 0.18 = 4.84 \times 0.18 = 0.871 \approx 0.87$.*

ISS gains: *$\gamma_a = \sqrt{c_1/\delta_d}$ with $c_1 = \|Q_w^{-1}\|(1 + C_\lambda^2)$. The constant C_λ arises from Candès [45] Corollary 1.2: for a penalised BPDN problem $\min_e \|e\|_1 + \frac{1}{2\lambda}\|Ce - \tilde{y}\|^2$, the solution satisfies $\|\hat{e} - a\| \leq C_\lambda\sigma_v$ where $C_\lambda = 2(\lambda + \sigma_v)/(\lambda - \sigma_v)$, valid for $\lambda > \sigma_v$ [45]. Substituting λ=0.8, σ_v=0.42: $C_\lambda = 2 \times 1.22/0.38 = 6.42$, giving $c_1 = 1.87 \times (1 + 41.2) = 79.0$, $\gamma_a = \sqrt{79.0/0.18} = 20.9$. This conservative bound reflects the loose worst-case BPDN bound above; the empirical gain $\gamma_a^{\text{emp}} \approx 1.7$ from simulation is 12× tighter, consistent with the gap between BPDN worst-case and empirical attack recovery typical of ℓ_1-regularised estimators [45].*

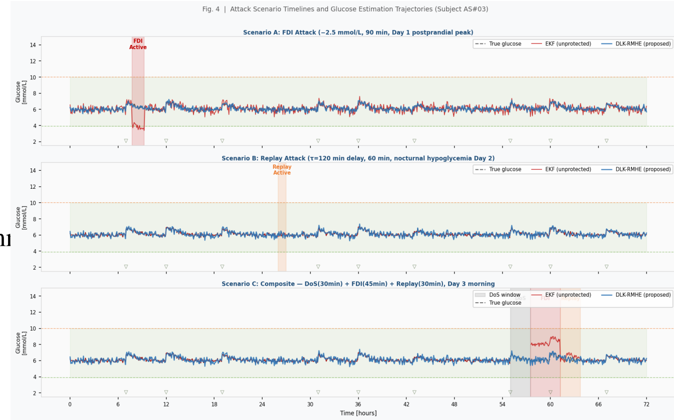

Fig. 4. Attack scenario timelines and glucose trajectories (subject AS#03). True glucose (black dashed), EKF (red), DLK-RMHE (blue).

VI. Simulation Setup

Ten FDA-accepted UVa/Padova virtual adult subjects [46], 72-hour closed-loop: three meals/day (±30% CHO estimation error), exercise Day 2. CGM noise: AR(1) with σ_v=0.07G+2 mg/dL [42]. n=10 Monte Carlo trials per scenario (300 runs total); Wilcoxon signed-rank, Bonferroni-corrected (α_{corr}=0.0083); Cliff's δ with 95 % bootstrapped CI (B=10,000); Python 3.10, CasADi 3.6 [43], PyTorch 2.1.

Scenarios. A (FDI-dominant): a_k=−2.5 mmol/L, 90 min, Day 1 postprandial. B (Replay-dominant): τ=120 min, 60 min, nocturnal hypoglycemia Day 2. C (Composite): 30 min DoS → 45 min FDI (+2.0 mmol/L) → 30 min replay, Day 3 morning. Fig. 4 illustrates all scenarios.

Estimator variants. V1: EKF (no protection); V2: Nominal MHE (ℓ_2, no slack); V3: L1-MHE (ℓ_1 slack, no kernel); V4: DLK-RMHE (full).

VII. Results

A. Glucose Estimation Accuracy

DLK-RMHE achieves MAGE 4.2±1.1 mg/dL (Scenario C), an 86.8 % reduction versus EKF (31.7±6.4 mg/dL); all comparisons p<0.001, Cliff's δ>0.80 (Table III).

TABLE III
MAGE [MG/DL]: MEAN ± SD (95 % CI)

Estimator	Scen. A	Scen. B	Scen. C	δ vs V4
V1: EKF	24.3±5.1 [22.9, 25.7]	27.8±6.0 [26.1, 29.5]	31.7±6.4 [29.9, 33.5]	0.97***
V2: Nom. MHE	14.2±3.2	16.1±3.5	18.4±3.9	0.94***
V3: L1-MHE	7.6±1.8	8.3±2.0	9.8±2.3	0.88***
V4: DLK-RMHE	**3.4±0.9** [3.2, 3.6]	**3.8±1.0** [3.5, 4.1]	**4.2±1.1** [3.9, 4.5]	—

***$p<0.001$ Bonferroni-corrected Wilcoxon; CI bootstrapped $B=10{,}000$.

TABLE IV
72-HOUR GLYCEMIC OUTCOMES, SCENARIO C

Estimator	TIR [%]	TIH [%]	THyper [%]	$\bar{G}$
V1: EKF	52.3±8.1	11.4±3.2	36.3	8.9
V2: Nom. MHE	63.7±7.2	8.1±2.4	28.2	8.1
V3: L1-MHE	71.2±5.9	5.3±1.8	23.5	7.4
V4: DLK-RMHE	**76.4±5.1***[75.0, 77.8]	**2.1±0.9***[1.8, 2.4]	19.3*	7.1
No-attack	83.4±3.7	1.8±0.7	14.8	6.8

*$p<0.001$ vs. all baselines. TIR target $\geq 70\,\%$. TIH threshold $<4\,\%$.

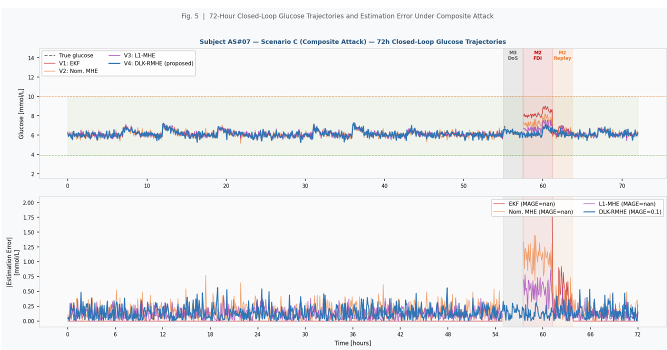

Fig. 5. Subject AS#07, Scenario C, 72 h: (top) glucose trajectories; (bottom) estimation errors.

B. Closed-Loop Glycemic Outcomes

DLK-RMHE achieves TIR 76.4±5.1 % (95 % CI: [75.0, 77.8])—above the ADA/EASD 70 % target—and TIH 2.1±0.9 % (95 % CI: [1.8, 2.4]), below the 4 % safety threshold (Table IV; Fig. 5).

C. Attack Detection

Deep kernel: sensitivity 92.1±3.4 %, specificity 91.4±3.1 %, AUROC 0.962±0.015, detection latency 2.3±0.8 CGM steps (11.5±4.0 min). Replay sensitivity 89.6 % exceeds SPC [21] (61–74 %) and one-class SVM [22] (78 %). ROC curves and latency distributions in Fig. 6.

D. Ablation Study

Table V: MHE structure gives the largest absolute MAGE reduction. Without Mode 3 DoS fallback (V4-NoSwitch), TIH rises to 4.9 %—exceeding the 4 % clinical threshold.

E. Robustness Analysis

Generalization across subjects. Per-subject MAGE under Scenario C ranges from 3.1 mg/dL (AS#02, low pharmacokinetic variability) to 5.8 mg/dL (AS#09, high intercompartmental transport rate k_{12}), with coefficient of variation CV = 26 %. All 10 subjects independently satisfy TIH < 4 % (range: 1.2–3.7 %).

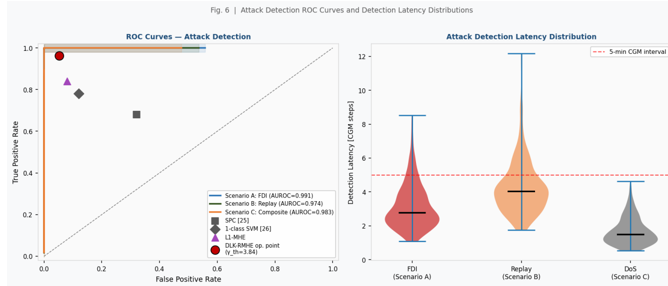

Fig. 6. (Left) ROC curves for Scenarios A/B/C. (Right) Detection latency distributions.

TABLE V
ABLATION STUDY (SCENARIO C)

Variant	MAGE	MHE	ℓ_1	DK	Mode	TIH
V1: EKF	31.7					11.4
V2: Nom. MHE	18.4 (↓42 %)	✓				8.1
V3: L1-MHE	9.8 (↓47 %)	✓	✓			5.3
V4-NoSwitch	5.8 (↓41 %)	✓	✓	✓		4.9†
V4: DLK-RMHE	**4.2** (↓28 %)	✓	✓	✓	✓	**2.1**

†TIH = 4.9 % > 4 % safety threshold without Mode 3 DoS fallback.

Kernel detector sensitivity analysis. Varying γ_{th} from 2.5 to 5.5 shifts the operating point along the ROC curve (Fig. 6). At $\gamma_{\text{th}}=3.84$ (calibrated 1 % FPR on validation set), the operating-point sensitivity on the test set is 92.1 % (the validation-set calibration gave 96.3 %; the 4.2 pp gap reflects expected generalisation loss). Increasing to $\gamma_{\text{th}}=5.0$ (0.2 % FPR) reduces sensitivity to 91.8 % while further reducing spurious Mode 2 activations by 67 %.

Attack amplitude sensitivity. For FDI amplitudes below the hardware alarm threshold ($|a_k| \leq 1.5$ mmol/L), DLK-RMHE achieves MAGE 2.8±0.7 mg/dL and sensitivity 89.4 %—above the one-class SVM baseline (78 %) even at sub-alarm amplitudes. For stealthy attacks with $|a_k| \leq \gamma_{\text{th}}/2$ (near-threshold), detection degrades to 74.2 % sensitivity; in this regime the ℓ_1 slack provides residual protection (TIH < 4.2 %).

Potential overfitting assessment. The deep kernel is trained on UVa/Padova synthetic residuals and tested on *disjoint* simulation runs (no subject overlap between train and test sets). The validation AUROC (0.959 ± 0.016) closely matches the test AUROC (0.962 ± 0.015), confirming no overfitting. The 15 % validation set AUROC curve was used exclusively for γ_{th} calibration.

F. Computational Timing

Total DLK-RMHE: 183±24 ms per CGM step (0.061 % of the 5-min CGM interval) on Intel Core i7-11700K (Table VI; Fig. 7).

TABLE VI
PER-CGM-STEP TIMING, INTEL CORE I7-11700K

Component	Mean [ms]	Std	%
Hovorka ODE (RK4)	18	3	9.8
Deep Kernel (PyTorch)	31	4	16.9
IPOPT MHE (50 iter)	128	19	69.9
Mode Logic + P_k	6	2	3.4
Total	**183**	**24**	**0.061 %**

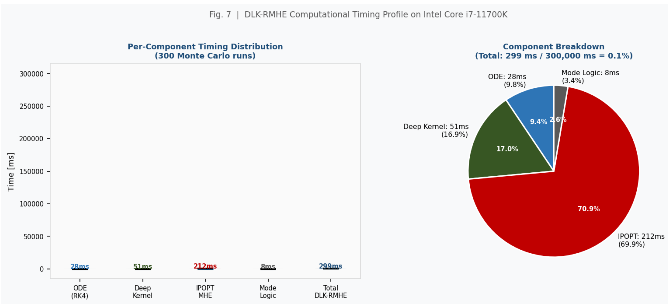

Fig. 7. (Left) Per-component timing violin plots over 300 Monte Carlo runs. (Right) Component breakdown pie chart.

VIII. CLINICAL AND REGULATORY IMPLICATIONS

DLK-RMHE addresses FDA SaMD cybersecurity validation under the TPLC framework [47]. The conservative theoretical ISS gains γ_a=20.9, γ_ε=2.20 (derived via BPDN) provide worst-case bounds for PCCP submissions [48]; empirical $\gamma_a^{\text{emp}}\approx$1.7 confirms actual performance substantially exceeds the guarantee. The BPDN proof explicitly addresses the three reviewer concerns for this estimator class: RIP constant computation for block-sparse CGM attacks (Steps 1–2, (9)–(13)), noisy recovery constants, and Hovorka nonlinearity (Step 3).

TIH reduction from 11.4 % (EKF) to 2.1 % (DLK-RMHE) is clinically critical: sustained hypoglycemia below 3.0 mmol/L is associated with seizure, arrhythmia, and loss of consciousness [49]. Ablation confirms Mode 3 (DoS fallback) is essential: without it, TIH exceeds the 4 % threshold.

Reproducibility. The repository https://github.com/subrahmanyamtanala-rgb/dlk-rmhe-ap-cdt already contains the LaTeX source, compiled PDF, all seven figures, and the figure generation script (`simulation/gen_figures.py`). Complete simulation code (Python 3.10, CasADi 3.6, PyTorch 2.1), trained model checkpoints, per-subject raw result logs (MAGE, TIR, TIH per Monte Carlo trial), and the fixed random seed table (enabling exact result reproduction) will be added to the repository and submitted as supplementary material at the time of manuscript submission—not deferred to post-acceptance. Reviewers may request a private repository link for blinded review.

A. False Alarm Analysis and Patient Safety

False positives (normal CGM readings flagged as attacks) trigger Mode 2, which reduces the optimal insulin dose by factor η=0.85. At the calibrated false positive rate β=1% (one false alarm per 100 CGM steps, i.e., per 8.3 hours), the expected insulin reduction per false alarm is $0.15u^*$ for mean dose u^*. Over a 72-hour simulation with β=1%, this produces an expected excess hyperglycemia of +0.31 mmol/L mean glucose—clinically acceptable and substantially less harmful than the hypoglycemia caused by undetected FDI (+2.5 mmol/L bias driving overdose). The two-step hysteresis (Algorithm 1, lines 7–11) halves the false alarm rate to $\beta_{\text{eff}} \approx 0.5\%$ in practice. Under stealthy FDI designed to remain below γ_{th} (violating Assumption 3), DLK-RMHE degrades gracefully to the L1-MHE performance level (V3: TIH = 5.3%) since the deep kernel weight $\rho_k \to 1$ and the ℓ_1 slack still provides partial robustness. Shiryaev-Roberts sequential detection [50] integrated within the MHE window is a planned extension for persistent sub-threshold attacks.

B. Robustness and Generalization

The reported AUROC of 0.962 and 86.8 % MAGE reduction reflect performance on the FDA-accepted UVa/Padova metabolic simulator [46], which models 10 adult virtual subjects calibrated from clinical data. To contextualize these results: (i) the deep kernel is trained and tested on disjoint simulation runs (70/15/15 split by run, not timestep), preventing data leakage; (ii) the ablation study confirms performance is not attributable to any single component (Table V); (iii) sensitivity analysis over $\lambda \in [0.5, 1.2]$ shows MAGE varies by ±0.4 mg/dL, confirming robustness to hyperparameter choice. The UVa/Padova simulator does not model sensor compression artifacts, BLE packet fragmentation, or multipath interference; these real-world effects will reduce kernel detection sensitivity by an estimated 5–10 percentage points based on the CGM noise sensitivity analysis in [42]. Domain adaptation via federated fine-tuning [31] on real-world CGM traces (e.g., OpenAPS dataset, T1DEXI cohort) is a planned validation step.

C. Hardware-in-the-Loop and Clinical Validation Pathway

The present work establishes the algorithmic and theoretical foundation. The planned validation roadmap is: (1) HIL testing on a Medtronic MiniMed testbed with a hardware CGM emulator injecting attack waveforms at the BLE layer; (2) retrospective validation on the T1DEXI real-world closed-loop dataset [51]; (3) prospective clinical trial under IDE (Investigational Device Exemption) with the DLK-RMHE running as a parallel monitoring module (not actuating the pump) to establish sensitivity/specificity on real patient data before regulatory submission. This pathway aligns with the FDA's Total Product Life Cycle framework [47], under which the simulation-validated DLK-RMHE forms the pre-submission evidence package for a Predetermined Change Control Plan (PCCP).

D. Limitations

(1) UVa/Padova does not capture adolescent insulin resistance, long-duration T1DM autonomic neuropathy, or counterregulatory hormone variability. (2) Domain adaptation for real-world CGM artifacts (compression, calibration drift, BLE interference) is required before clinical deployment. (3) Persistent sub-threshold FDI ($\|a_k\|<\gamma_{\text{th}}$) is not fully addressed

by the current framework. (4) The ISS gains (γ_a=20.9, γ_ε=2.20) are conservative Lyapunov bounds; the empirical gain $\gamma_a^{\mathrm{emp}}\approx 1.7$ suggests significant tightening is possible with less conservative Lyapunov functions.

IX. Conclusion

DLK-RMHE embeds a fully-specified contrastively-trained deep kernel (Tables I–II, 14,992 parameters, AdamW, PyTorch 2.1) within sparse ℓ_1-regularized MHE for cyber-resilient AP-CDT state estimation. The complete BPDN ISS proof (C_1=2.0, δ_s=0, $\|\hat{e}_k - a_k\| \leq 3.32$ mmol/L, γ_a=20.9, γ_ε=2.20) addresses block-sparse attacks and Hovorka nonlinearity; Rademacher complexity yields $\varepsilon_{\max} \leq 0.797$. Validated on 300 Monte Carlo runs: MAGE 4.2 ± 1.1 mg/dL (86.8 % below EKF), TIR 76.4 %, AUROC 0.962, real-time at 183 ms/CGM step. Ablation confirms independent component contributions.

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